

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_random_portfolio.

Run Metadata

- run_name: run_20260520_164657_b
- started_at_local: 2026-05-20 16:46:57
- finished_at_local: 2026-05-20 17:26:44
- duration_hhmm: 00:40
- duration_seconds: 2386.581
- seed_offset: 1000
- replicate_label: b
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1472.465143	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.18258	0.11032	445.345514	0.18258	0.085372	0.121185	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1483.437971	0.072798	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1563.561671	0.07673	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.214311	0.142051	471.443086	0.214311	0.053836	0.112958	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.084733	0.012473	4200.622143	0.241725	0.021201	0.044607	0.0	0.62	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.120261	0.048001	10916.496586	0.891586	0.090132	0.101232	0.181687	0.69	False
phase8_full_solver_evolution	full_solver	none	0.203742	0.131482	291.650157	0.203742	0.011221	0.08215	0.0	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1472.465143 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.18258, selector regret 0.11032, runtime-adjusted gap 0.18258.
- Family transfer 0.085372 and combined transfer 0.121185.
- Runtime 445.345514 ms, runtime inflation -0.697551, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072798.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1483.437971 ms, runtime inflation 0.007452, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07673.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1563.561671 ms, runtime inflation 0.061867, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.053836 and combined transfer 0.112958.
- Runtime 471.443086 ms, runtime inflation -0.679827, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:limited_three_opt:candidate_pruned_two_opt:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-conditioned selector over a frozen portfolio; picks backbone + refinement family based on structure descriptors (grid/bottleneck/cluster/corridor/trap) with conservative shared schedule tuning knobs.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "tsp_hh_static_interpretable_selector_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "candidate_pruned_two_opt", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.084733, selector regret 0.012473, runtime-adjusted gap 0.241725.
- Family transfer 0.021201 and combined transfer 0.044607.
- Runtime 4200.622143 ms, runtime inflation 1.852782, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:nearest_neighbor_multistart:stag3:cand2:restart2.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "candidate_pruned_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller selecting from a frozen TSP heuristic portfolio with bounded stagnation-triggered schedule tuning.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "cheapest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "tsp_hh_evolved_adaptive_controller_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "candidate_pruned_two_opt", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

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"tsphh_portfolio_interpretable_v1", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt",
"random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic":
"nearest_neighbor_multistart", "stagnation_threshold": 3, "temperature_scale": 1.12, "time_budget_trigger": 0.55,
"two_cluster_bottleneck_heuristic": "edge_preserving_restart"}
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phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.891586.
- Family transfer 0.090132 and combined transfer 0.101232.
- Runtime 10916.496586 ms, runtime inflation 6.413756, mean novelty 0.181687, and complexity 0.69.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:nearest_neighbor_multistart:stag2:cand0:restart2.
- Controller rule summary: {"acceptance_bias": 0.01, "candidate_limit_offset": 0, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller using frozen heuristic portfolio. Structure-gated backbone selection (cluster/grid/bottleneck/corridor/trap) with stagnation-aware switching and bounded schedule tuning to improve held-out transfer.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.06, "name": "tsp_hh_oracle_portfolio_interpretable_epoch4_deterministic", "nearest_neighbor_trap_heuristic": "cheapest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 2, "temperature_scale": 1.15, "time_budget_trigger": 0.45, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.203742, selector regret 0.131482, runtime-adjusted gap 0.203742.
- Family transfer 0.011221 and combined transfer 0.08215.
- Runtime 291.650157 ms, runtime inflation -0.801931, mean novelty 0.0, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

Primary Endpoint: Held-out TSPLIB Gap

- ****Best Single Fixed Heuristic ("farthest_insertion_two_opt")**** achieves the lowest held-out TSPLIB gap: ****0.07226****.
- The ****Oracle Selector**** matches this gap at ****0.07226****, serving as an unattainable upper bound.
- Other adaptive or learned selectors (e.g., supervised ML, LLM static/adaptive controllers) have notably worse gaps (≥ 0.0767 and up to 0.2143), failing to improve upon the best fixed heuristic baseline on held-out TSPLIB.

Selector Regret vs. Oracle Portfolio

- The best fixed heuristic has zero selector regret (by definition).
- The supervised ML selector also reaches zero regret, closely matching the best heuristic.

- Adaptive controllers show some selector regret (e.g., 0.0125 for evolved adaptive controller, 0.048 for replay), indicating they mispredict or inefficiently select heuristics relative to the oracle.
- The static and random portfolio selectors have higher selector regret (≥ 0.11), reflecting less effective adaptation.

Runtime-Adjusted Gap & Runtime Inflation

- The best single fixed heuristic baseline has no runtime inflation and moderate runtime-adjusted gap (0.0723).
- The oracle selector nearly matches this runtime-adjusted gap with negligible inflation (+0.75%).
- The supervised ML selector incurs slight runtime inflation (~6%) with slightly worse gap.
- The adaptive controllers incur substantial runtime inflation (e.g., 185% for evolved adaptive, 641% for replay), with larger runtime-adjusted gaps (0.24 to 0.89), indicating expensive overheads that diminish practical benefit.
- The static LLM selector and random portfolio have negative runtime inflation (faster than baseline) but much worsened runtime-adjusted gaps due to poorer solution quality.

Pareto Efficiency

- Pareto-efficient conditions (in terms of gap and complexity) are:
- Best Single Fixed Heuristic
- Full Solver Evolution (gap 0.20, runtime-inflation -80%)
- Random Portfolio (gap 0.18, runtime-inflation -70%)
- Adaptive selectors are **not** Pareto efficient here, due to high runtime inflation and modest quality improvements.

Cross-Family Transfer (Transfer Gap)

- Best fixed and supervised selectors have transfer gaps ~ 0.031 , indicating moderate robustness across instance families.
- Adaptive controllers show higher but still moderate transfer gaps (0.04 – 0.10).
- Oracle selector transfer gap is lowest (0.027), as expected.
- The random and static selectors show the largest transfer gaps (0.11 – 0.12), consistent with poorer generalization.

Summary & Recommendations

- The **best single fixed heuristic ("farthest_insertion_two_opt")** provides a strong, interpretable baseline with excellent held-out TSPLIB gap, zero selector regret, no runtime inflation, and Pareto efficiency.
- The **oracle selector confirms this baseline is near-optimal**, justifying conservative claims that high adaptivity or hyper-heuristic control has limited room to improve held-out gap.
- **Supervised ML selector matches held-out performance closely but with mild runtime inflation**; it demonstrates that learning-based control can approach best fixed heuristic performance without excessive overhead, supporting interpretable instance-adaptive selection.
- **Adaptive controllers deliver moderate quality improvements within family but incur large runtime overheads and higher selector regret**, undermining practical benefits on held-out TSPLIB; careful tuning needed before deployment.
- **Static selectors and random portfolios are generally inferior**, with larger gaps and higher regret; they offer less interpretable or instance-adaptive control.

- This phase supports hyper-heuristic control focused on **interpretable, bounded adaptation over known heuristics**, rather than discovery of new heuristics.
- Further work should prioritize **runtime-efficient adaptive control that closely matches or improves on the best fixed heuristic in held-out benchmarks** before claims of deployment-ready adaptivity.

In conclusion, the current portfolio suite's best static heuristic and supervised selectors show promise for interpretable instance-adaptive control with conservative runtime cost. More aggressive adaptation, while improving some metrics, introduces excessive runtime inflation and selector regret on held-out TSPLIB tasks, limiting practical gains.